International Parkinson and Movement Disorder Society
Meeting Abstracts

Incidence and Prevalence of Parkinson's Disease Among United States Medicare Beneficiaries by Age, Sex, Race/Ethnicity, and Neighborhood Socioeconomic Status 2014-2021

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Objective: To evaluate the trends and differences in incidence and prevalence of Parkison's disease (PD) in Medicare beneficiaries between 2014-2021 across different demographics.

Background: The prevalence of PD is rising, resulting in greater economic and healthcare burden. In the United States, few studies have evaluated how demographic and economic factors are associated with incidence and prevalence of PD diagnosis nationally. Our study uses one of the largest samples to date to demonstrate significant differences in the incidence and prevalence of Medicare beneficiaries with PD by age, sex, race and economic status.

Method: Data from the Centers for Medicare and Medicaid Services (2013-2021) were linked with Area Deprivation Index (ADI; a measure of neighborhood economic status). Medicare beneficiaries with PD were identified using validated ICD-9-CM or ICD-10-CM codes. Incidence and prevalence were reported over time and by age, sex, race/ethnicity, and ADI.

Results: A total of 1,133,207 beneficiaries (Incident PD mean age 78.5) were included. The incidence rates of PD decreased, and the prevalence rate increased over the study period [Table 1]. The incidence were highest for white (88.6%), male individuals (55.3%), and those in the South (40.5%) region [Figure 1]. Medicare beneficiaries living in the highest ADI deciles (representing low neighborhood SES) had the lowest PD incidence and prevalence [Table 2]. Black beneficiaries with PD were more likely to be living in the greatest ADI decile and were threefold more likely than White individuals to be dually eligible for Medicaid. Black individuals were more likely to be

diagnosed with dementia (65.7% Black, 59.7% White) and psychosis (20.4% Black, 13.2% White). Beneficiaries with incident PD at the greatest ADI decile had the highest proportion of female (49.2%) and black patients (16.1%), a higher likelihood to be dual Medicaid eligible (29.1%), and the majority were located on the South (58.2%).

Conclusion: The differences in incidence and prevalence rates across sex, race and area deprivation levels suggests that the interplay between biological, social and economic index is associated with PD diagnosis. Our findings can help policy makers and payers make informed decisions for equitable resource allocation.

Variable	2014	2015	2016	2017	2018	2019	2020	2021
N person-years	24,134,760	24,169,360	24,520,553	24,686,218	24,694,808	24,611,406	24,430,128	23,680,764
Incident	188,161 (7.80)	163,104 (6.75)	152,555 (6.22)	146,978 (5.95)	144,552 (5.85)	140,498 (5.71)	119,384 (4.89)	131,462 (5.55)
Prevalent	641,234 (24.98)	678,698 (26.26)	716,805 (27.21)	732,778 (27.80)	744,374 (28.30)	756,801 (28.76)	715,583 (27.75)	702,624 (28.21)

Table 1. Incident and prevalence rates of Parkinson's disease in Medicare FFS beneficiaries, 2014-2021, per 10,000 person-years

Variable	ADI 1st Decile	ADI 2nd Decile	ADI 3rd Decile	ADI 4th Decile	ADI 5th Decile	ADI 6th Decile	ADI 7th Decile	ADI 8th Decile	ADI 9th Decile	ADI 10th Decile	p-value
N	129,468	125,443	130,653	132,342	125,686	119,058	110,047	100,882	92,511	67,117	
Neurodegenerative subgroups											
Dementia	76,534 (59.1%)	75,247 (60.0%)	78,104 (59.8%)	78,911 (59.6%)	75,329 (59.9%)	71,686 (60.2%)	66,041 (60.0%)	60,350 (59.8%)	56,115 (60.7%)	41,125 (61.3%)	< .001
Parkinson's Disease	129,468 (100.0%)	125,443 (100.0%)	130,653 (100.0%)	132,342 (100.0%)	125,686 (100.0%)	119,058 (100.0%)	110,047 (100.0%)	100,882 (100.0%)	92,511 (100.0%)	67,117 (100.0%)	
Alzheimer's Disease	14,456 (11.2%)	13,729 (10.9%)	13,667 (10.5%)	13,553 (10.2%)	12,886 (10.3%)	12,389 (10.4%)	11,396 (10.4%)	10,900 (10.8%)	10,438 (11.3%)	8,042 (12.0%)	< .001
Other dementia	67,563 (52.2%)	66,753 (53.2%)	69,818 (53.4%)	70,952 (53.6%)	67,853 (54.0%)	64,641 (54.3%)	59,521 (54.1%)	54,175 (53.7%)	50,172 (54.2%)	36,367 (54.2%)	< .001
Demographics											
Age, years, Mean (SD)	79.5 (7.5)	78.9 (7.4)	78.7 (7.4)	78.5 (7.3)	78.4 (7.4)	78.4 (7.4)	78.3 (7.4)	78.2 (7.4)	78.0 (7.4)	77.8 (7.5)	< .001
Age categories											< .001
65-69	13,158 (10.2%)	14,537 (11.6%)	15,969 (12.2%)	16,996 (12.8%)	16,263 (12.9%)	15,966 (13.4%)	15,206 (13.8%)	13,988 (13.9%)	13,679 (14.8%)	10,683 (15.9%)	
70-74	23,456 (18.1%)	24,305 (19.4%)	25,954 (19.9%)	26,671 (20.2%)	25,402 (20.2%)	23,939 (20.1%)	22,116 (20.1%)	20,601 (20.4%)	18,895 (20.4%)	14,188 (21.1%)	
75-79	28,711 (22.2%)	28,320 (22.6%)	30,048 (23.0%)	30,697 (23.2%)	29,080 (23.1%)	27,341 (23.0%)	25,101 (22.8%)	23,072 (22.9%)	21,122 (22.8%)	15,116 (22.5%)	
80-84	28,751 (22.2%)	27,153 (21.6%)	28,203 (21.6%)	28,143 (21.3%)	26,872 (21.4%)	25,432 (21.4%)	23,396 (21.3%)	21,397 (21.2%)	19,412 (21.0%)	13,551 (20.2%)	
85-89	22,408 (17.3%)	20,379 (16.2%)	20,040 (15.3%)	19,962 (15.1%)	18,682 (14.9%)	17,544 (14.7%)	16,314 (14.8%)	14,771 (14.6%)	13,058 (14.1%)	9,119 (13.6%)	
90+	12,984 (10.0%)	10,749 (8.6%)	10,439 (8.0%)	9,873 (7.5%)	9,387 (7.5%)	8,836 (7.4%)	7,914 (7.2%)	7,053 (7.0%)	6,345 (6.9%)	4,460 (6.6%)	
Sex, Female	54,991 (42.5%)	54,155 (43.2%)	56,268 (43.1%)	56,914 (43.0%)	55,537 (44.2%)	53,723 (45.1%)	50,801 (46.2%)	46,993 (46.6%)	44,063 (47.6%)	32,990 (49.2%)	< .001
Race/Ethnicity											< .001
White	109,661 (84.7%)	107,688 (85.8%)	116,193 (88.9%)	120,439 (91.0%)	114,906 (91.4%)	109,197 (91.7%)	100,357 (91.2%)	91,386 (90.6%)	81,702 (88.3%)	52,869 (78.8%)	
Black	3,601 (2.8%)	5,705 (4.5%)	5,480 (4.2%)	5,475 (4.1%)	5,342 (4.3%)	5,486 (4.6%)	5,823 (5.3%)	6,035 (6.0%)	7,462 (8.1%)	10,802 (16.1%)	
Other	16,206 (12.5%)	12,050 (9.6%)	8,980 (6.9%)	6,428 (4.9%)	5,438 (4.3%)	4,375 (3.7%)	3,867 (3.5%)	3,461 (3.4%)	3,347 (3.6%)	3,446 (5.1%)	
Dual Medicaid Eligibility	18,264 (14.1%)	18,257 (14.6%)	15,503 (11.9%)	14,129 (10.7%)	14,377 (11.4%)	15,770 (13.2%)	16,559 (15.0%)	17,686 (17.5%)	20,115 (21.7%)	19,508 (29.1%)	< .001
Region											< .001
Northeast	40,691 (31.4%)	40,656 (32.4%)	34,343 (26.3%)	28,119 (21.2%)	21,636 (17.2%)	16,739 (14.1%)	13,315 (12.1%)	9,529 (9.4%)	7,201 (7.8%)	4,660 (6.9%)	
South	23,778 (18.4%)	36,394 (29.0%)	45,568 (34.9%)	52,247 (39.5%)	53,167 (42.3%)	53,783 (45.2%)	52,226 (47.5%)	51,132 (50.7%)	51,481 (55.6%)	39,080 (58.2%)	
Midwest	5,559 (4.3%)	11,567 (9.2%)	20,234 (15.5%)	27,081 (20.5%)	32,646 (26.0%)	35,533 (29.8%)	35,276 (32.1%)	33,616 (33.3%)	29,076 (31.4%)	19,292 (28.7%)	
West	59,421 (45.9%)	36,795 (29.3%)	30,470 (23.3%)	24,818 (18.8%)	18,161 (14.4%)	12,862 (10.8%)	9,054 (8.2%)	6,366 (6.3%)	4,319 (4.7%)	3,617 (5.4%)	
Other	19 (0.0%)	31 (0.0%)	38 (0.0%)	77 (0.1%)	76 (0.1%)	141 (0.1%)	176 (0.2%)	239 (0.2%)	434 (0.5%)	468 (0.7%)	
Comorbidities											
Anxiety disorders	32,765 (25.3%)	34,607 (27.6%)	37,350 (28.6%)	38,902 (29.4%)	37,326 (29.7%)	36,176 (30.4%)	34,366 (31.2%)	31,625 (31.3%)	29,727 (32.1%)	21,732 (32.4%)	< .001
Cancer (metastatic or non-metastatic)	26,250 (20.3%)	24,040 (19.2%)	24,118 (18.5%)	23,959 (18.1%)	22,067 (17.6%)	19,875 (16.7%)	17,812 (16.2%)	16,006 (15.9%)	14,222 (15.4%)	10,237 (15.3%)	< .001
Cerebrovascular disease	48,259 (37.3%)	47,200 (37.6%)	48,680 (37.3%)	48,734 (36.8%)	46,855 (37.3%)	44,290 (37.2%)	41,328 (37.6%)	37,856 (37.5%)	35,161 (38.0%)	26,376 (39.3%)	< .001
Chronic pulmonary disease	32,193 (24.9%)	34,458 (27.5%)	37,047 (28.4%)	38,534 (29.1%)	38,882 (30.9%)	38,521 (32.4%)	36,936 (33.6%)	35,566 (35.3%)	34,076 (36.8%)	26,183 (39.0%)	< .001
Congestive heart failure	26,820 (20.7%)	27,007 (21.5%)	28,098 (21.5%)	28,899 (21.8%)	28,886 (23.0%)	28,376 (23.8%)	27,349 (24.9%)	26,528 (26.3%)	25,913 (28.0%)	19,829 (29.5%)	< .001
Depression	45,432 (35.1%)	46,131 (36.8%)	48,988 (37.5%)	49,929 (37.7%)	47,809 (38.0%)	45,358 (38.1%)	41,468 (37.7%)	38,017 (37.7%)	35,365 (38.2%)	25,431 (37.9%)	< .001
Diabetes mellitus	43,742 (33.8%)	45,610 (36.4%)	46,717 (35.8%)	47,785 (36.1%)	46,829 (37.3%)	46,102 (38.7%)	43,884 (39.9%)	41,729 (41.4%)	39,727 (42.9%)	31,176 (46.5%)	< .001
Hypertension	102,925 (79.5%)	102,730 (81.9%)	107,690 (82.4%)	109,635 (82.8%)	105,457 (83.9%)	100,901 (84.7%)	94,102 (85.5%)	87,274 (86.5%)	80,766 (87.3%)	59,707 (89.0%)	< .001
Coronary heart disease	48,802 (37.7%)	48,915 (39.0%)	50,506 (38.7%)	51,690 (39.1%)	50,084 (39.8%)	48,285 (40.6%)	45,566 (41.4%)	42,896 (42.5%)	39,937 (43.2%)	29,878 (44.5%)	< .001
Myocardial infarction	12,086 (9.3%)	12,510 (10.0%)	13,545 (10.4%)	14,291 (10.8%)	14,295 (11.4%)	13,563 (11.4%)	13,047 (11.9%)	12,419 (12.3%)	11,605 (12.5%)	8,689 (12.9%)	< .001
Peripheral vascular disease	45,432 (35.1%)	46,131 (36.8%)	48,988 (37.5%)	49,929 (37.7%)	47,809 (38.0%)	45,358 (38.1%)	41,468 (37.7%)	38,017 (37.7%)	35,365 (38.2%)	25,431 (37.9%)	< .001
Psychoses	16,373 (12.6%)	16,887 (13.5%)	16,699 (12.8%)	17,060 (12.9%)	16,280 (13.0%)	16,030 (13.5%)	15,314 (13.9%)	14,129 (14.0%)	13,832 (15.0%)	10,919 (16.3%)	< .001
Renal disease	28,828 (22.3%)	29,988 (23.9%)	32,127 (24.6%)	33,260 (25.1%)	32,875 (26.2%)	31,951 (26.8%)	30,001 (27.3%)	28,089 (27.8%)	26,653 (28.8%)	20,209 (30.1%)	< .001
Stroke/TIA	37,008 (28.6%)	36,363 (29.0%)	37,318 (28.6%)	37,472 (28.3%)	36,115 (28.7%)	34,116 (28.7%)	32,024 (29.1%)	29,305 (29.0%)	27,280 (29.5%)	20,656 (30.8%)	< .001
Valvular heart disease	32,169 (24.8%)	31,064 (24.8%)	31,807 (24.3%)	31,365 (23.7%)	30,087 (23.9%)	28,116 (23.6%)	25,956 (23.6%)	23,883 (23.7%)	21,431 (23.2%)	15,769 (23.5%)	< .001
County Resources per 1000 residents											
Total MDs 2020, Median (Q1, Q3)	18.6 (14.4, 23.1)	15.8 (13.5, 21.1)	14.9 (11.8, 18.6)	13.9 (10.6, 18.0)	13.3 (9.9, 17.4)	12.4 (8.8, 16.2)	11.4 (8.0, 15.9)	10.5 (7.3, 14.9)	9.7 (6.6, 14.2)	10.1 (6.9, 14.6)	< .001
Primary Care Physicians 2020, Median (Q1, Q3)	18.6 (14.4, 23.1)	15.8 (13.5, 21.1)	14.9 (11.8, 18.6)	13.9 (10.6, 18.0)	13.3 (9.9, 17.4)	12.4 (8.8, 16.2)	11.4 (8.0, 15.9)	10.5 (7.3, 14.9)	9.7 (6.6, 14.2)	10.1 (6.9, 14.6)	< .001
NPs and PAs 2020, Median (Q1, Q3)	11.0 (8.6, 14.9)	13.1 (9.4, 16.1)	13.3 (9.5, 17.0)	12.9 (9.7, 17.4)	12.9 (9.5, 17.7)	12.6 (9.0, 17.7)	12.3 (8.7, 17.4)	11.9 (8.1, 16.9)	11.4 (7.9, 16.5)	11.8 (8.4, 16.9)	< .001
Neurologists 2020, Median (Q1, Q3)	0.6 (0.5, 1.0)	0.6 (0.4, 0.9)	0.5 (0.3, 0.8)	0.4 (0.2, 0.7)	0.4 (0.2, 0.7)	0.3 (0.1, 0.6)	0.2 (0.0, 0.6)	0.2 (0.0, 0.5)	0.2 (0.0, 0.4)	0.2 (0.0, 0.5)	< .001
Psychiatrists 2020, Median (Q1, Q3)	1.6 (1.4, 2.3)	1.5 (0.9, 1.8)	1.1 (0.6, 1.7)	0.9 (0.5, 1.5)	0.7 (0.4, 1.3)	0.6 (0.3, 1.1)	0.5 (0.2, 1.0)	0.4 (0.0, 0.9)	0.3 (0.0, 0.8)	0.4 (0.0, 0.8)	< .001
Hospital Beds 2020, Median (Q1, Q3)	23.4 (19.1, 30.1)	23.9 (18.7, 31.8)	23.9 (17.4, 32.9)	25.0 (16.5, 34.1)	26.0 (16.5, 35.8)	26.2 (16.1, 37.8)	25.8 (15.2, 38.9)	25.7 (14.7, 39.2)	25.7 (14.3, 40.2)	27.5 (17.0, 41.3)	< .001

Table 2. Baseline characteristics of Medicare FFS beneficiaries with incident Parkinson's disease from 2014-2021, by ADI decile

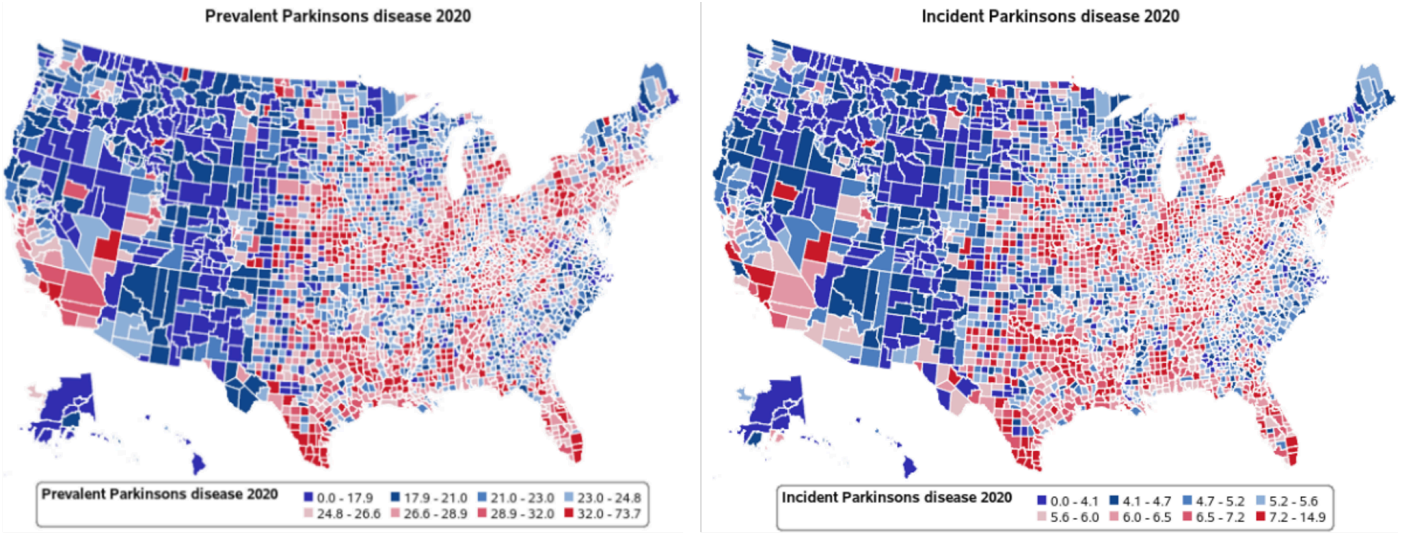

Figure 1. Incident and Prevalent PD (2020)